

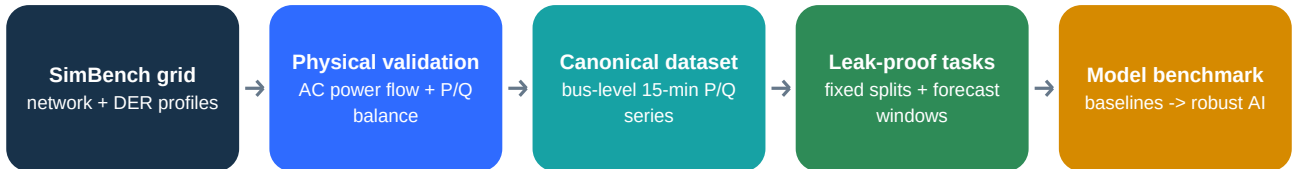
OpenGridFlex / GridShiftBench

What we have built so far, and what the benchmark is designed to prove

Goal

Build a trustworthy benchmark for AI forecasting in active distribution grids - with reproducible data, physical validation, strict anti-leakage rules, and later robustness testing under distribution shift.

The benchmark in one picture



What makes this different from a normal forecasting script?

1. Physics first

Before any ML, transformer settings, profile mappings, units, signs, AC convergence and system power balance are checked.

2. No hidden leakage

The full timeline is chronological. Scaling uses training data only, and forecast targets cannot cross split boundaries.

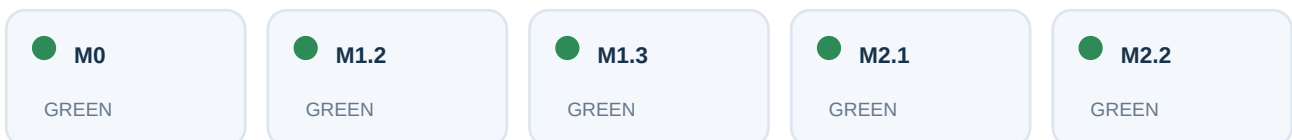
3. Reproducible by design

Pinned software versions, fingerprints, manifests, hashes, deterministic seeds and regression gates make every stage auditable.

4. Grid-aware outputs

The core targets are bus-level net active and reactive demand, so forecasts can later be checked through AC power flow.

Current status



M3.1 has been implemented as the first baseline layer; its gate should be run and frozen before moving to learned models.

What exactly is the benchmark doing?

GridShiftBench turns one validated distribution grid year into fixed forecasting tasks that every model must solve under the same rules.

1. Canonical physical dataset

Item	Frozen definition
Grid	SimBench 1-MV-urban--1-no_sw
Resolution	15-minute physical timeline
Length	35,136 timesteps
Spatial level	136 buses
Core channels	load P/Q, sgen P, gen P, storage P, net-demand P/Q
Net-demand sign	load + storage - sgen - gen; positive = net consumption
Time handling	UTC canonical timeline; local calendar/DST kept as known-future features

2. Frozen forecasting tasks

Primary - gsb_bus_pq_4h_v1

Use the previous 24 h (96 steps) to forecast the next 4 h (16 steps). Targets: bus-level net P and Q.

Secondary - gsb_bus_pq_24h_v1

Use the previous 7 physical days (672 steps) to forecast the next 24 h (96 steps). Targets: the same bus-level net P and Q.

3. Train / validation / test discipline

Split	Local period	Purpose
Train	Jan 1 -> Sep 13	Fit model + preprocessing
Validation	Sep 13 -> Nov 7	Develop and compare methods
Test	Nov 7 -> Jan 1	Final holdout; should stay sealed during development

Anti-leakage rule: every forecast can only use measurements that were available at its forecast origin. Any target horizon crossing a split boundary is discarded. Normalization statistics are fitted on the training period only.

What has already been verified?

Each stage has a fail-closed gate. A stage only becomes GREEN when its tests, scientific checks and upstream regression checks pass.

Stage	What it guarantees	Status
M0	Engineering foundation: configs, tests, seeds, provenance, lint/CI.	GREEN
M1.2	SimBench/grid integrity: profile structure, DST-safe time, transformer tap repair, AC regression.	GREEN
M1.3	Physical reconstruction: relative profiles -> MW/Mvar -> network -> AC PF; P/Q balance verified.	GREEN
M2.1	Canonical bus dataset + chronological windows + train-only normalization + leakage audit.	GREEN
M2.2	Frozen benchmark tasks, calendar-aligned splits, metrics, quantile grid, serialization + hashes.	GREEN
M3.1	Three non-learned references: last value, 24 h seasonal, 7 d seasonal. Validation only; test sealed.	CURRENT

How models will be judged

Deterministic accuracy

P and Q are reported separately using MAE, RMSE and WAPE.

Probabilistic quality

Later quantile models use pinball loss plus 90% interval coverage, width and coverage error.

Multiple aggregation views

Performance is tracked at bus-micro, active-bus macro, system aggregate and each horizon step.

Future robustness layer

Later milestones add distribution shifts: season/weather, DER penetration, topology, sensor loss and unseen feeders.

Why this matters for the paper

The benchmark is not only asking, "Which model has the lowest error?" It is building evidence that a model remains accurate, calibrated and physically useful when the operating conditions of a distribution grid change. That makes the later contribution much stronger than a standard single-dataset forecasting comparison.

Immediate next step

Run and freeze M3.1. If the three sanity baselines pass, M3.2 can introduce the first learned tree-based deterministic/quantile baseline. Every learned model must use the exact same M2.2 task, split and evaluation contract.